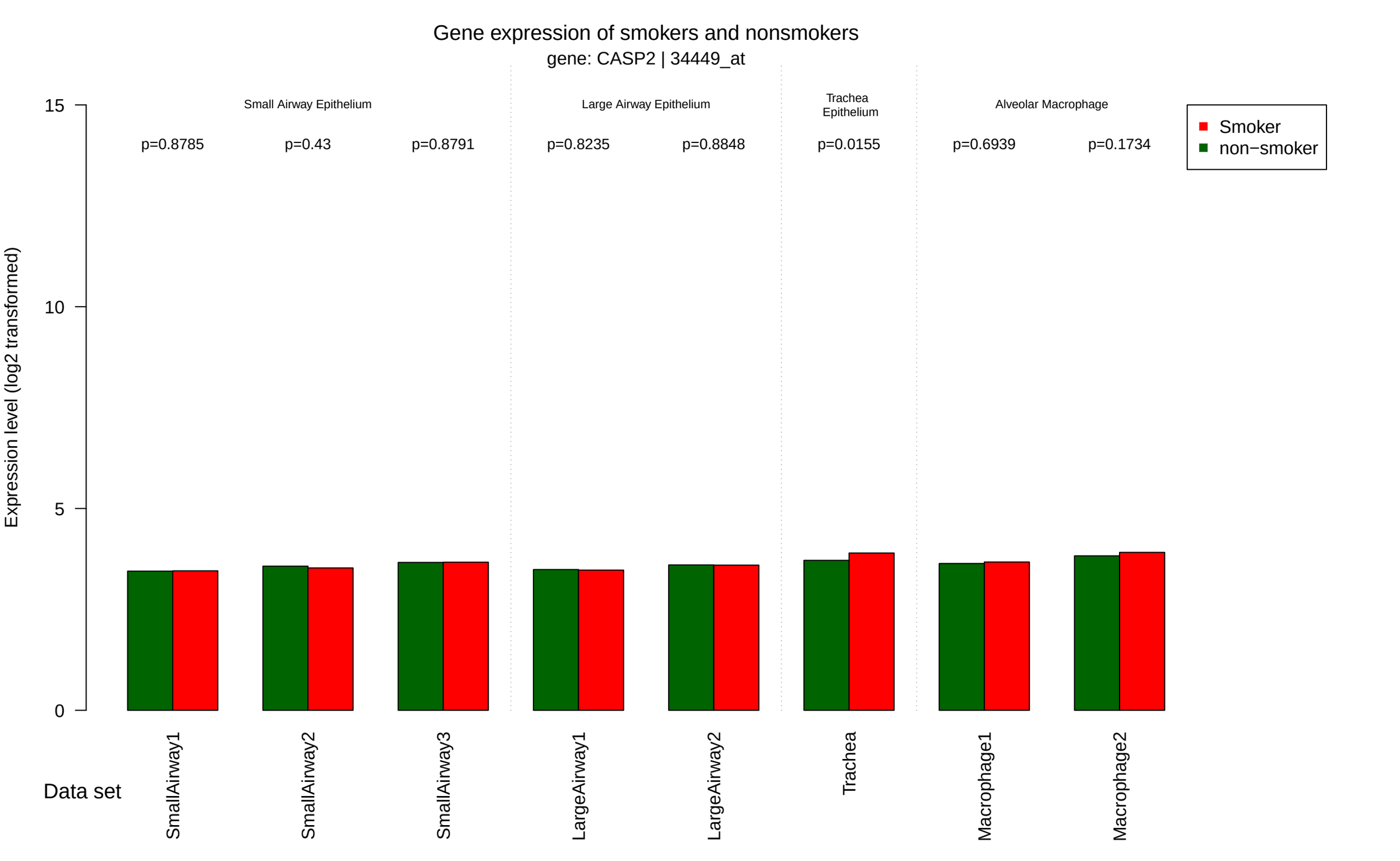
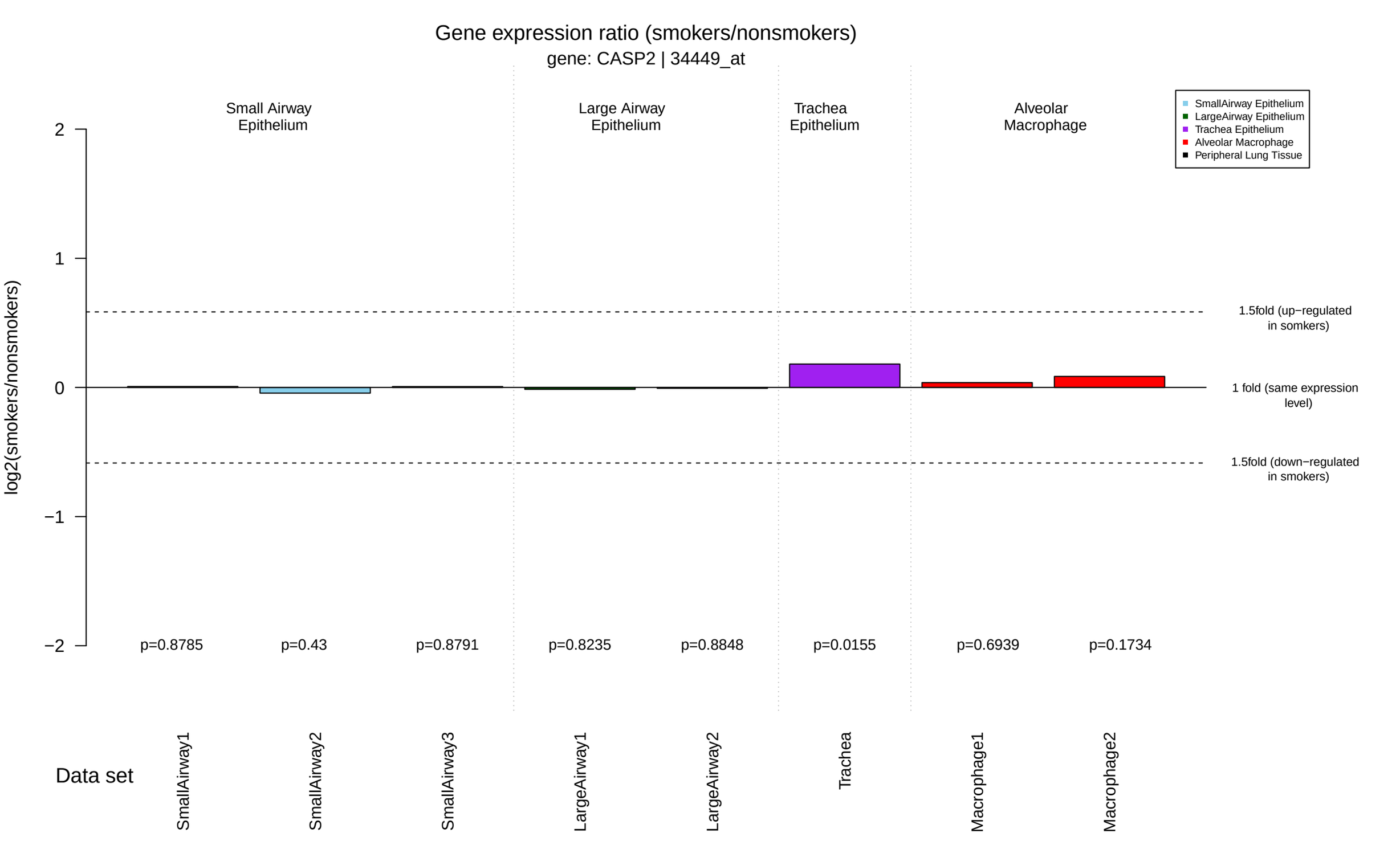
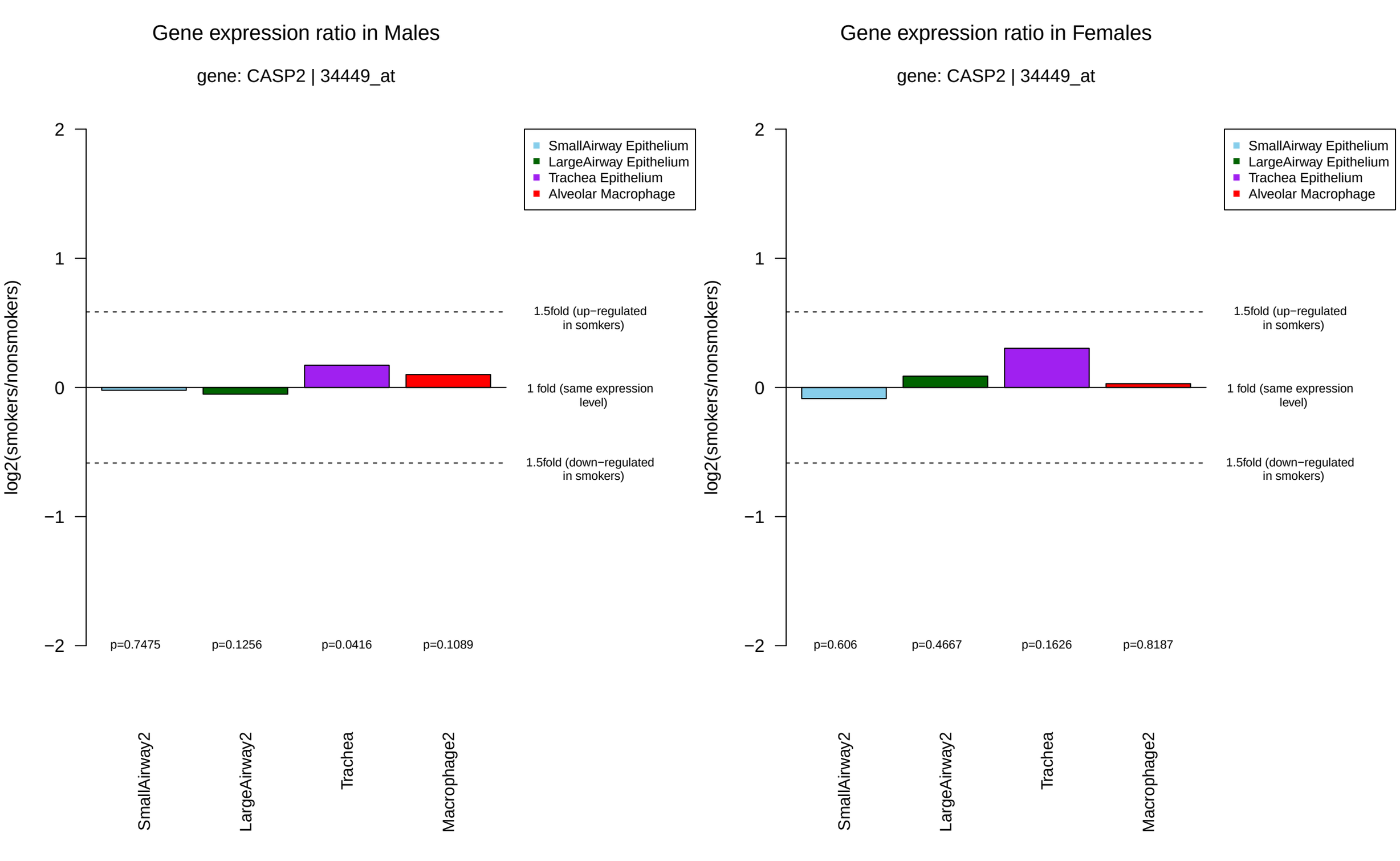
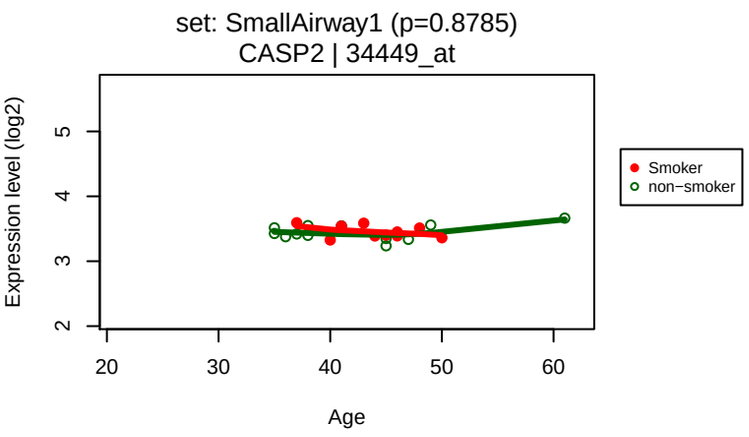
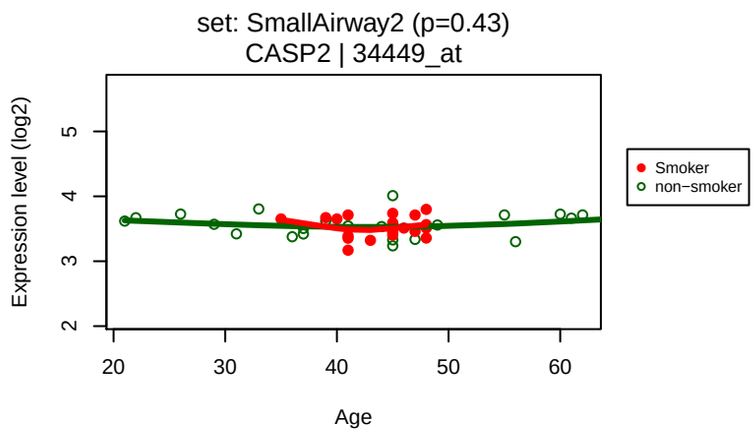
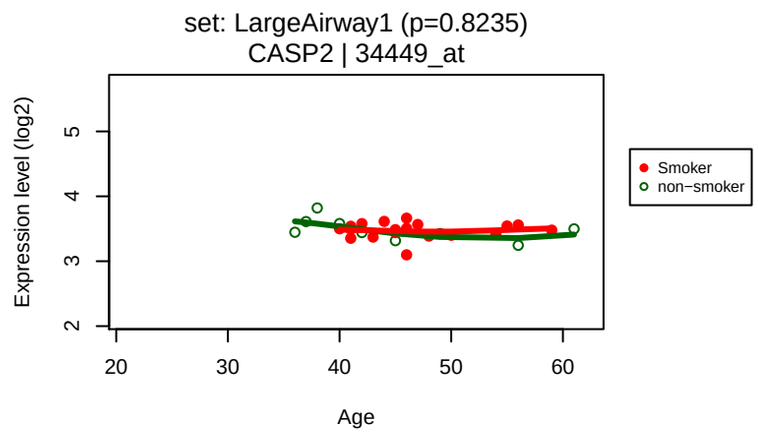

Gene expression of smokers and nonsmokers
gene: CASP2 | 34449_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8785
p=0.43
p=0.8791
p=0.8235
p=0.8848
p=0.0155
p=0.6939
p=0.1734
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

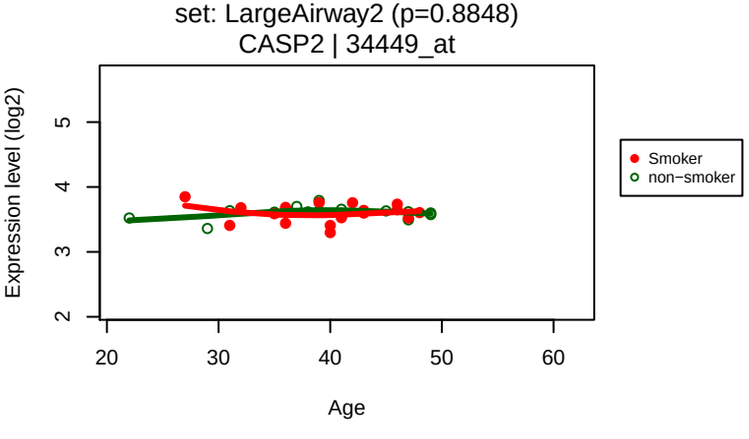

Gene expression ratio (smokers/nonsmokers)
gene: CASP2 | 34449_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8785
p=0.43
p=0.8791
p=0.8235
p=0.8848
p=0.0155
p=0.6939
p=0.1734
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

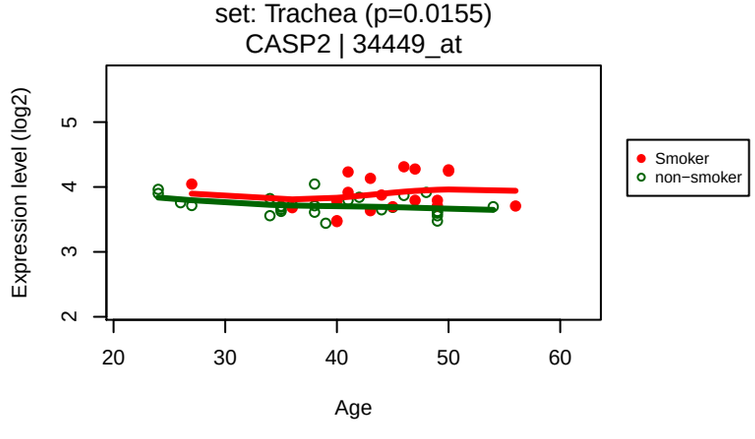

Gene expression ratio in Males
gene: CASP2 | 34449_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7475
p=0.1256
p=0.0416
p=0.1089
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CASP2 | 34449_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.606
p=0.4667
p=0.1626
p=0.8187
SmallAirway2
LargeAirway2
Trachea
Macrophage2

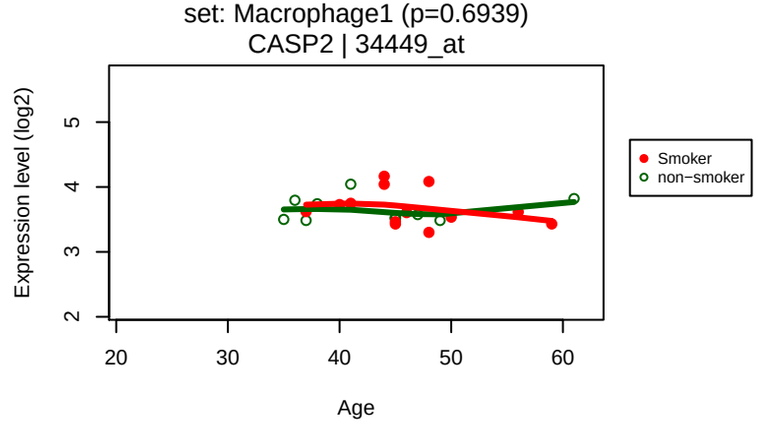

set: SmallAirway1 (p=0.8785)
CASP2 | 34449_at
Expression level (log2)
Age
Smoker
non−smoker

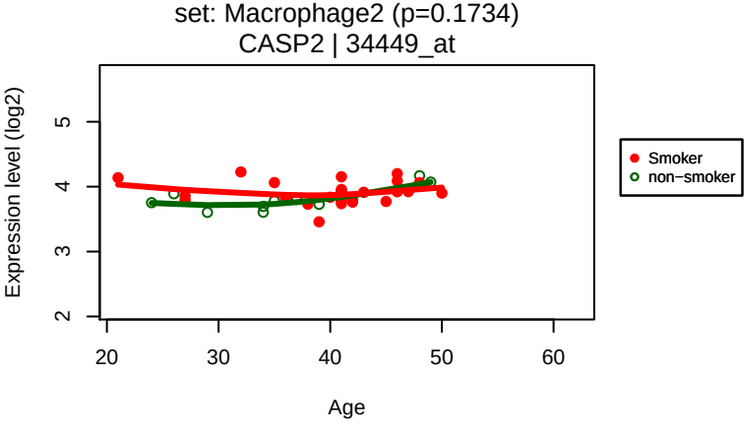

set: SmallAirway2 (p=0.43)
CASP2 | 34449_at
Expression level (log2)
Age
Smoker
non−smoker

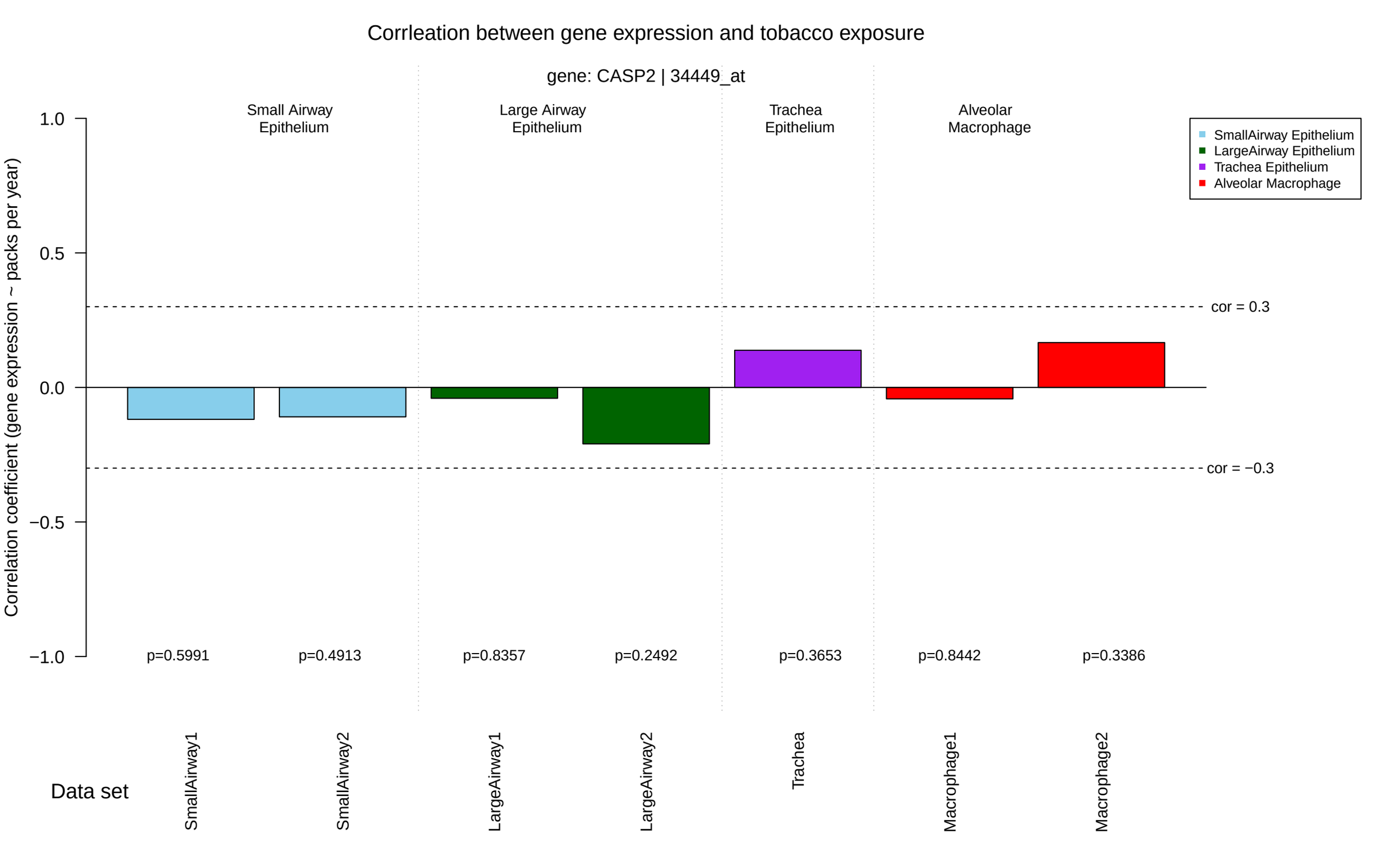

set: LargeAirway1 (p=0.8235)
CASP2 | 34449_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8848)
CASP2 | 34449_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0155)
CASP2 | 34449_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6939)
CASP2 | 34449_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1734)
CASP2 | 34449_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CASP2 | 34449_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5991
p=0.4913
p=0.8357
p=0.2492
p=0.3653
p=0.8442
p=0.3386
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2